

Case study: Reconstruction of infection dynamics during a pandemic using imperfect surveillance and serial serological survey data

- During **infectious disease outbreak**, public health authorities often rely on **multiple sources of information to monitor disease spread** and **evaluate the successfulness of intervention strategies**
- In the case study, **participants will analyze simulated outbreak data** from a pandemic occurring in a closed structured population

NIAS **Lorentz center**

Connecting Survival Analysis and Infectious Disease Modeling

15 - 19 June 2026, Leiden, the Netherlands @lambda

1: Alive event-free $\lambda_{12}(t|X)$ **2: Illness Covid-19 Infection** $\lambda_{23}(t|X)$ **3: Outcome Death** $\lambda_{13}(t|X)$

Topics

- Survival analysis methods (multi-state models, frailty models, additive hazard models, competing risks)
- Infectious disease modelling (compartmental SIR/SEIRD models, reproduction number estimation)
- Estimating vaccine efficacy and time-dependent vaccine effectiveness
- Heterogeneity in susceptibility and infectiousness
- Handling incomplete data: censoring, reporting delays, and underreporting

Scientific Organizers

- Liesbeth de Wreede, Leiden University Medical Center (LUMC)
- Hein Putter, LUMC
- Kylie Ainslie, University of Melbourne and University of Hong Kong
- Don Klinkenberg, National Institute for Public Health and the Environment (RIVM) and Wageningen University & Research
- Jacco Wallinga, RIVM and LUMC
- Steven Abrams, UHasselt and University of Antwerp

This workshop is part of the NIAS-Lorentz Program, which brings together perspectives from humanities & social sciences with natural & technological sciences.

LORENTZCENTER.NL

Universiteit Leiden, Universiteit Wageningen, UHasselt, ZonMw, NWO

S(usceptible)I(nfected)R(ecovered)V(accinated)D(eath) IBM

- Simulated data is based on the application of an **individual-based microsimulation model** (HERMES) in a structured population comprising **households, schools, workplaces**, and the **general community**.
- An **S(usceptible)I(nfected)R(ecovered)V(accinated)D(eath)**-structure with individuals navigating between the different health states is adopted.
- Before starting simulation of transmission events, HERMES constructs a **synthetic population of individuals for which demographic data are available**.
- Transmission events are simulated explicitly for each infected individual on each day, i.e., every infectious individual can generate **new infections in their household, the community, school, and/or workplace**.
- Transmission is governed by **setting-specific effective contact probabilities** that translate into **transmission probabilities**.

Unfolding of the outbreak

- The outbreak starts at day 0 with the introduction of disease in the target **population of size $N = 200,000$ individuals**
- The **first 20 days**, there is **unrestricted spread of the disease** in the population after which a **lockdown is imposed**, leading to contact reductions at the workplace, in schools and in the general community.
- At **day 35** an **exit** out of the lockdown is present, leading to contact probabilities increasing to pre-lockdown values.
- At **day 45**, part of the population will start to get vaccinated with **increasing vaccination coverage over three distinct cross-sectional time points**.

Datasets

1. Demographic data
2. Incidence data (cases and hospitalizations)
3. Serial cross-sectional serological data
4. Contact tracing data
5. Vaccination data
6. Mortality data

Objectives of the case study

1. Estimate key epidemiological parameters, including the basic/effective reproduction number, recovery time, incubation period, and reporting delay distribution.
2. Derive the serial interval distribution based on the available contact tracing data.
3. Estimate parameters related to the reporting-delay distribution and study underreporting in the incidence data through the use of serial serological survey data to estimate the proportion of underreporting.
4. Estimate vaccine effectiveness using individual-level vaccination history and breakthrough infection data.
5. Estimate excess mortality and its relation with infection and vaccination.

Additional details in overview document

Case study: Reconstruction of infection dynamics during a pandemic using imperfect surveillance and serial serological survey data

Background:

During an emerging infectious disease outbreak, public health authorities often rely on multiple sources of information to monitor disease spread and evaluate the successfulness of intervention strategies. Reported case counts provide information on the number of detected infections, while serological surveys on the other hand are useful to provide information on the cumulative proportion of the population that has detectable antibodies related to infection in the past, thereby including undetected cases in the incidence data.

In this case study, participants will analyze simulated outbreak data from a pandemic occurring in a closed structured population. The simulation model incorporates infection transmission, vaccination, disease progression, and mortality. Next to the available demographic data, several other datasets are available representing different sources of information that may be collected during a real-world disease outbreak.